

Lecture 14

Good Data, Bad Outcomes II

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"After analyzing all your data, I think we can safely say that none of it is useful."

Last Time

- ❖ Ways analyses could go wrong even with clean data
 - Survivorship Bias
 - Selection bias
 - Goodhart's Law

Today

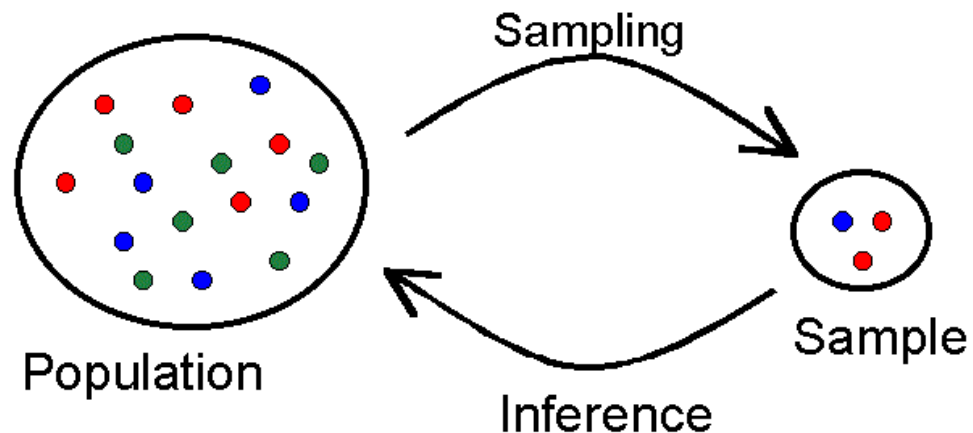
Ways analyses can go wrong even with clean data

- ❖ Confounding
- ❖ Simpson's Paradox

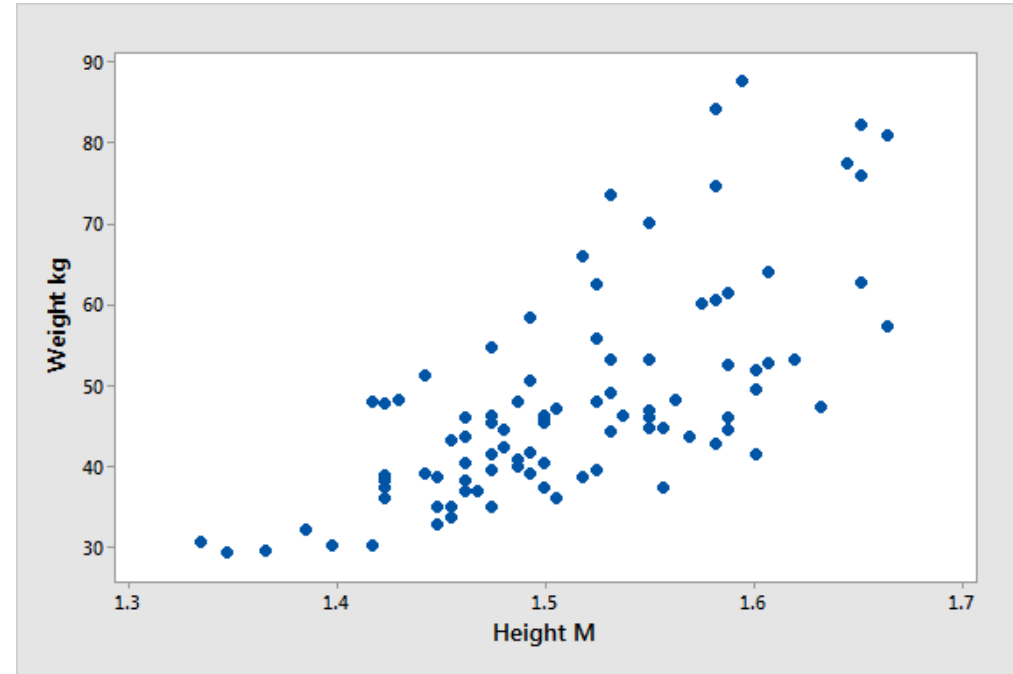
Look at datasets

Inference

- ❖ Drawing conclusions about the world from our dataset
- ❖ An inference is valid if what's going on in the sample corresponds to what's going on the population
- ❖ Validity is determined by
 - Methods used to reason about data
 - Provenance of our data



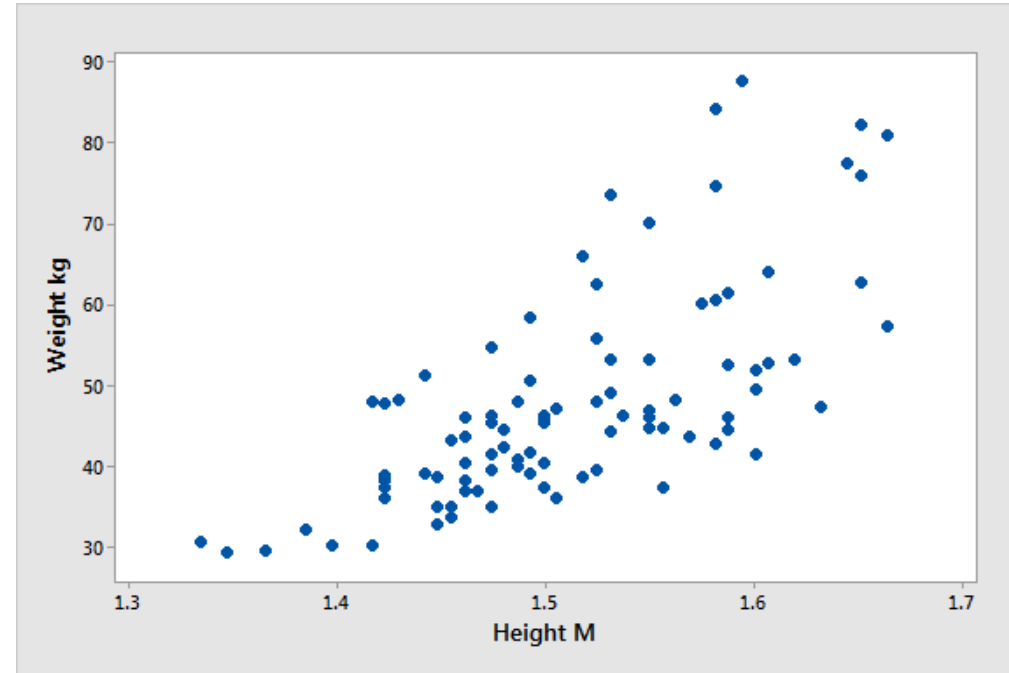
Correlations & Hypotheses



Correlations & Hypotheses

When forming hypotheses, we often develop a **mechanism**

- ❖ A process for how the data is generated
- ❖ Typically involves causation



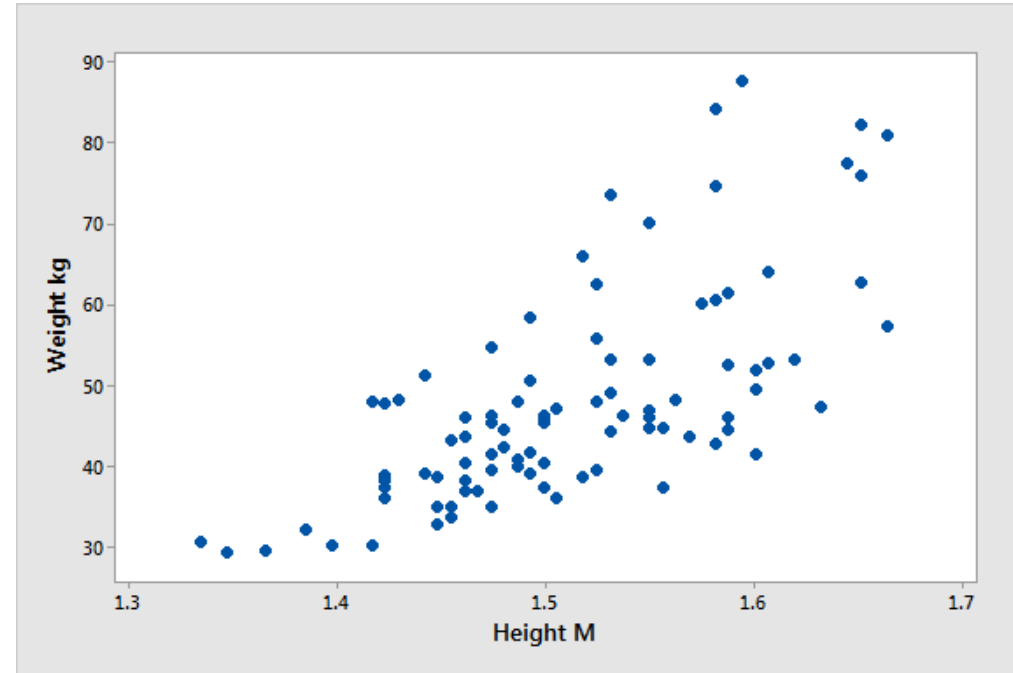
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Example:

- ❖ Taller people are just bigger on average
- ❖ Positing a biological mechanism



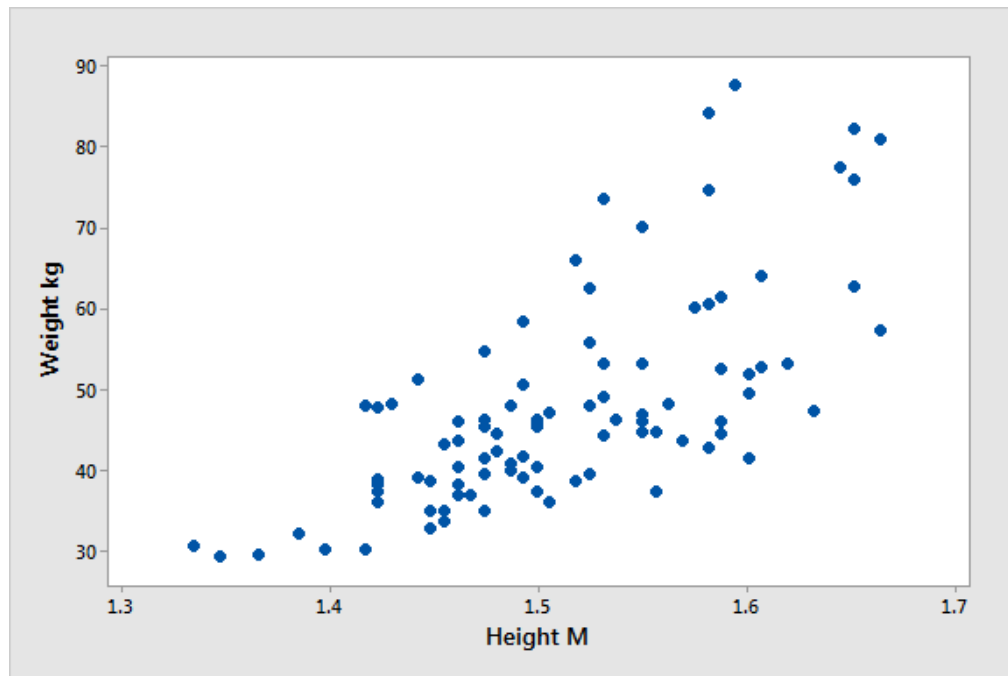
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- ❖ Taller people are just bigger on average
- ❖ Positing a biological mechanism
- ❖ "telling a story"



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Correlations can be misleading

- ❖ We may posit false mechanisms to explain them
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Coffee & Heart Attacks

- ❖ Researchers in the 1950s observed a strong positive correlation between coffee consumption and heart attacks
- ❖ Posited that caffeine consumption from coffee was causing heart attacks
- ❖ Reasonable mechanism that would explain the data

Mechanisms

What makes a mechanism reasonable or convincing?

Mechanisms

What makes a mechanism reasonable or convincing?

- ❖ Does it explain the phenomena in the data?
- ❖ Does it cohere with other things we know?

Confounders

A correlation between two variable may be driven by a "common cause"

Coffee & Heart Attacks

- ❖ Smoking is a confounder
- ❖ **Controlling** for smoking, the correlation between coffee and heart attacks vanishes

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- ❖ For a binary confounder, we can measure correlation among each group separately

Simpson's Paradox

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- ❖ A trend appears in several different groups of data, but disappears or reverses when those groups are combined

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- ❖ Example: batting averages in baseball

Batter \ Year	1995		1996		Combined	
Derek Jeter	12/48	.250	183/582	.314	195/630	.310
David Justice	104/411	.253	45/140	.321	149/551	.270

Next Time

- ❖ Start visualizations
- ❖ Introduce course project
 - Deliverables and due dates
 - Walk through topics (each team will work on separate project)
 - Release teams afterward